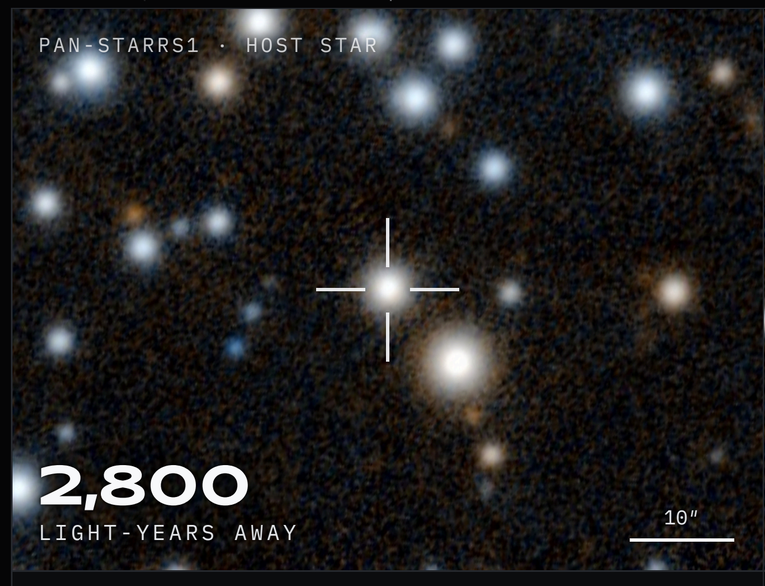
THREE SUB-NEPTUNE PLANET CANDIDATES

# THREE NEW WORLDS IN THE MILKY WAY

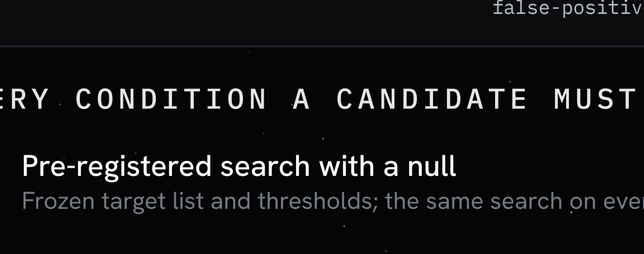
Found by Aayush Gupta in NASA Kepler data from 2009–2013, around faint stars the telescope recorded but never targeted, 2,800 to 6,400 light-years away. Each passes every test the data allow, with a false-positive probability below 1 %.[1, 2]

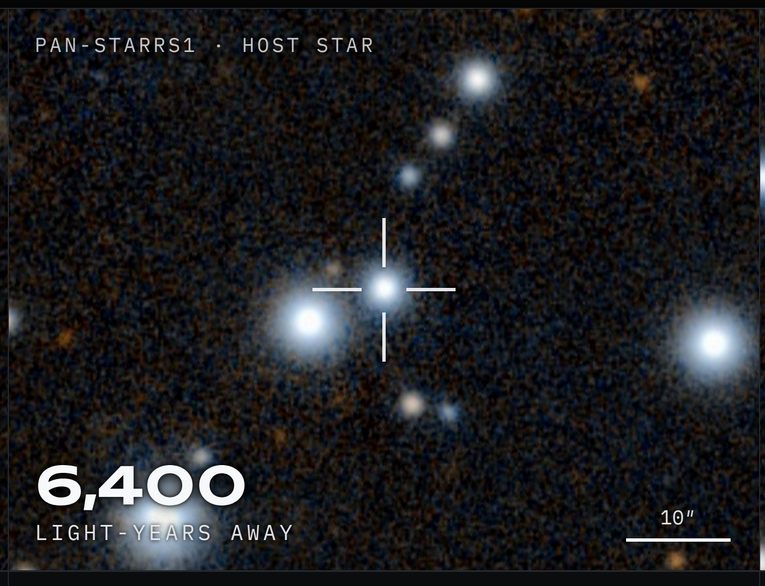
## THE THREE HOST STARS, PHOTOGRAPHED, AND THE TRANSITS THAT BETRAY THEIR CANDIDATES




### KIC 6699887 — 0.63 %

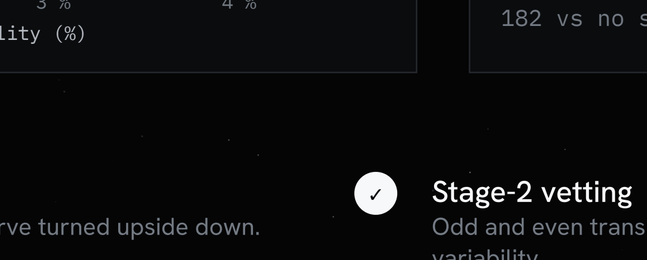
15.1136-day orbit · 2.76 Earth radii
1,023 ppm · 86 transits · $T_{eq}$ ≈ 611 K
host 4,542 K dwarf, G 16.2, 859 pc
FPP ~~2.4 %~~ → 0.63 % with Gaia limits
second on-target signal at 47.69 d

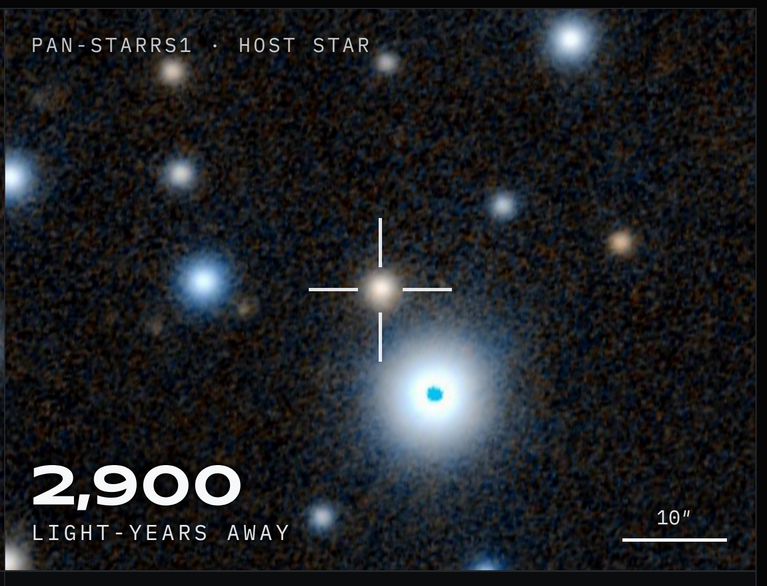



### KIC 3831969 — 0.71 %

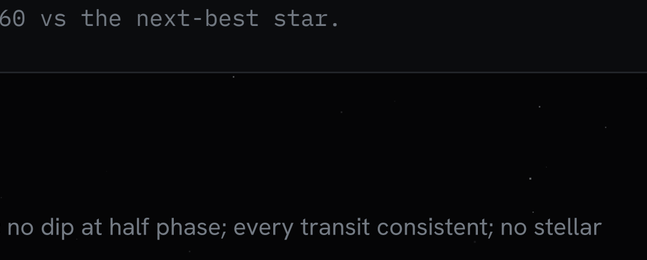
15.3133-day orbit · 3.47 Earth radii
997 ppm · 86 transits · $T_{eq}$ ≈ 722 K
host 5,275 K dwarf, G 16.5, 1,969 pc
FPP ~~2.7 %~~ → 0.71 % with Gaia limits




### KIC 11138423 — 0.70 %

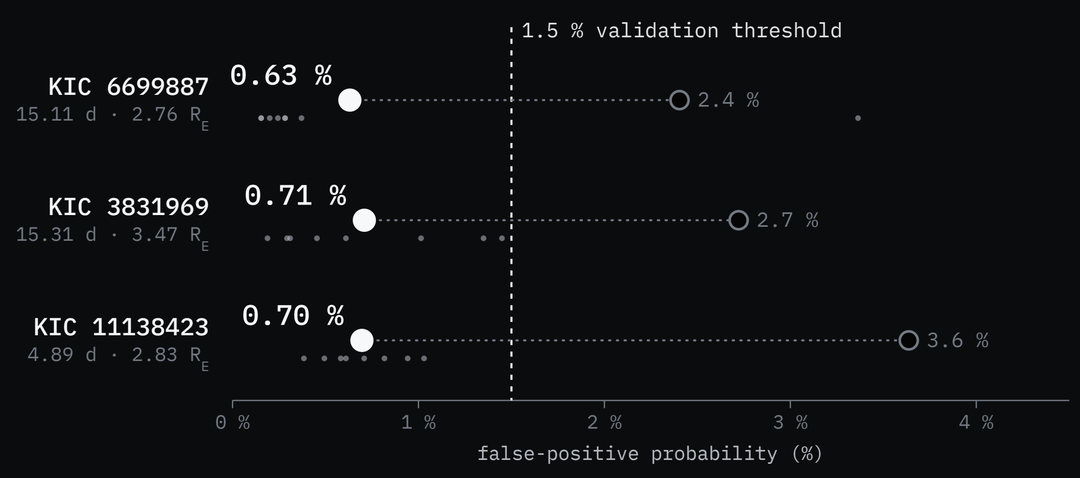
4.8945-day orbit · 2.83 Earth radii
1,316 ppm · 253 transits · $T_{eq}$ ≈ 720 K
host 4,258 K dwarf, G 17.4, 894 pc
FPP ~~3.6 %~~ → 0.70 % with Gaia limits



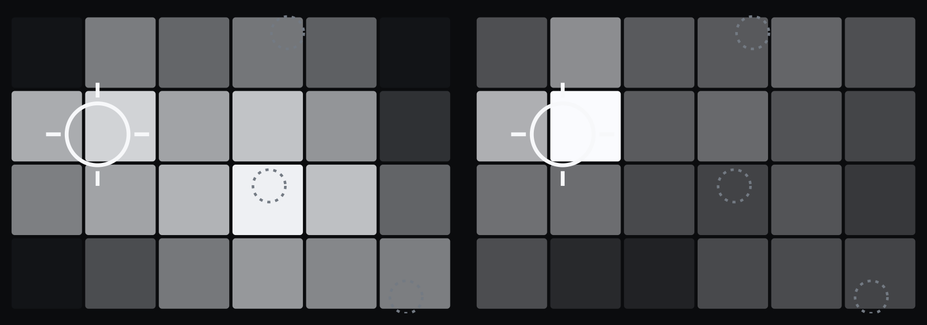

## HOW THEY WERE FOUND

| 6,465 | 448 | 22 | 5 | 4 | 3 |
|---|---|---|---|---|---|
| stars searched, thresholds frozen first | first-screen passes (upside-down null: 97) | pass stage-2 vetting | lose light on their own pixels | survive re-derivation and red team | candidates, FPP below 1 % |

## FALSE-POSITIVE PROBABILITY




## THE LIGHT GOES MISSING FROM THE TARGET



KIC 6699887, Kepler season 2. Left: the stamp of the bright star Kepler observed. Right: in-transit minus out-of-transit light. The missing light sits on the candidate's Gaia position (ring): $\Delta\chi^2$ 182 vs no signal, 60 vs the next-best star.

## EVERY CONDITION A CANDIDATE MUST MEET

- ✓ **Pre-registered search with a null**
  Frozen target list and thresholds; the same search on every light curve turned upside down.
- ✓ **Light lost on the target's own pixels**
  Two independent pixel tests, per quarter and per season.
- ✓ **Independent re-derivation**
  A separate agent with no shared code recovered period and depth.
- ✓ **Transit shape fits the host star**
  Stellar density from the transit matches Gaia within 0.5 σ.
- ✓ **Stage-2 vetting**
  Odd and even transits match; no dip at half phase; every transit consistent; no stellar variability.
- ✓ **No known object shares the ephemeris**
  Checked against every KOI, threshold-crossing event and eclipsing binary in the Kepler field.
- ✓ **Red-team review**
  Agents tasked with killing each signal found no false-positive mechanism.
- ✓ **False-positive probability below 1.5 %**
  TRICERATOPS with Gaia constraints: 0.63, 0.71 and 0.70 %; nearby stars 0 %.

### CANDIDATE

A periodic dip that survives every test the data allow: localised on the star, absent from the inverted-flux null, unmatched by any known object, and unlikely to be anything else.

### VALIDATED

False-positive probability below 1.5 % and nearby-star probability below 0.1 % in a refereed paper, usually with a high-resolution image of the host. Not yet done here.

### CREDITS

Kepler Bonus photometry (Martínez-Palomera et al. 2023; doi:10.17909/7jbr-w430), Gaia DR3, TRICERATOPS, MOLUSC. Sky images: Pan-STARRS1 via CDS hips2fits, 1.2′ fields, north up; the candidates themselves are not resolved.

[1] Data & code: doi:10.5281/zenodo.23442225

[2] Research Notes of the AAS, AAS82486 (submitted)